This question paper contains 4 printed pages. **Your Roll No.** ....

Sl. No. of Ques. Paper	**: 5709**	**F**
Unique Paper Code	**: 234501**	
Name of Paper	**: Theory of Computation**	
Name of Course	**: B.Sc. (Hons.) Computer Science**	
Semester	**: V**	
Duration	**: 3 hours**	
Maximum Marks	**: 75**	

(Write your Roll No. on the top immediately on receipt of this question paper.)

NOTE : *Assume $\Sigma = \{a, b\}$ is the underlying alphabet unless mentioned otherwise.*

All questions from Part A are compulsory. Attempt any four questions from Part B. Parts of a question must be answered together.

Part A

Q1

(a) Compare Deterministic Finite Automaton and Transition Graph. (2)

(b) Describe the language associated with regular expression $(a(aa)^*b(bb)^*)^*$. (3)

(c) Prove that the set of Regular Languages is closed under complement. (3)

(d) State Pumping Lemma for Regular Languages. Is the language $L=\{ (bab)^i(babbab)^i : i\geq 1 \}$ defined over an alphabet $\Sigma=\{a,b\}$ regular ? Justify your answer. (4)

(e) Give regular expression for the language L where $L = \{ a^n b^m \mid m>=0, n>=3\}$ (4)

(f) Prove that a language is recursive if and only if both it and its complement are recursively enumerable. (4)

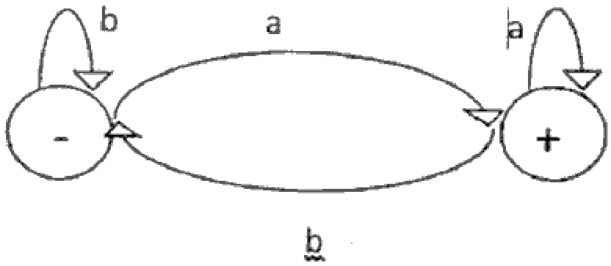
(g) Write Context Free Grammar for the language: (5)

$$L = \{ w \in \{a,b\}^* : w \text{ has twice as many b's as a's} \}$$

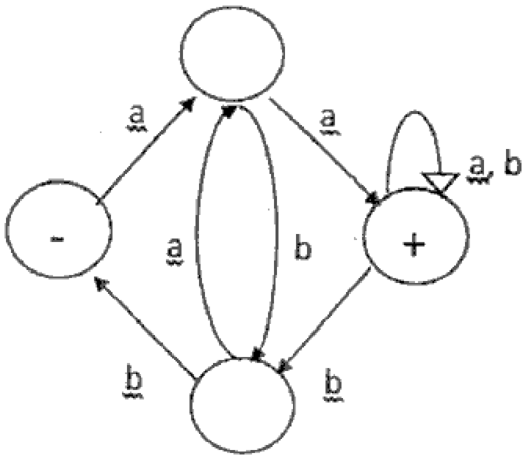
(h) Describe the operation of Universal Turing Machine. (5)

(i) Given the following machines: (5)

Turn over



FA1



FA2

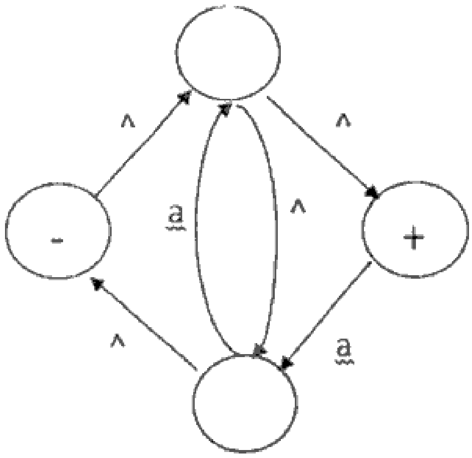
Determine union of these two machines i.e. FA1 + FA2.

Part B

Q2

(a) Rewrite the regular expression $(a+b)^* ab(a+b)^* + b^* a^*$ as simpler regular expression. (2)

(b) Convert the following Non-Deterministic Finite Automaton (NFA) into Deterministic Finite Automaton (FA) (4)



(c) Describe the Kleene Closure FA^* for the following Finite Automaton (FA) (4)

Q3 Given the regular expressions $(b+ab)^*(a+\wedge)$ and $(a+b)^*aa(a+b)^*$ corresponding to languages L1 and L2 respectively, determine a regular expression and an FA that define L1∩L2. (10)

Turn over

Q4

(a) Show that the following context free grammar (CFG) is ambiguous : (4)

S ->aS | bS | aaS | ^

(b) Find Context Free Grammar for the language defined over an alphabet { a , b } consisting of all strings without the substring aaa. (6)

Q5

(a) Is the Language L = { $a^n b^n c^n$: n = 1, 2, 3, . . . } context free ? Justify. (4)

(b) Give Push Down Automaton (PDA) for the language : (6)
L = { $a^n b^{n+1}$; n = 1,2,3, . . . }.

Q6

(a) Build a Turing machine for the language : (3)

L = { $a^n b^n : n \geq 0$ }

(b) Define k-tape Turing machine. How a configuration of such machine is expressed? (3)

(c) Design Turing machine for a language defined over an alphabet { a , b } where each string consists of at least one a. (4)

Q7

(a) Define recursive function in context of Turing Machines. (3)

(b) Design 2 tape Turing machine for performing addition of two binary numbers. (7)

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